

Combinations & Permutations

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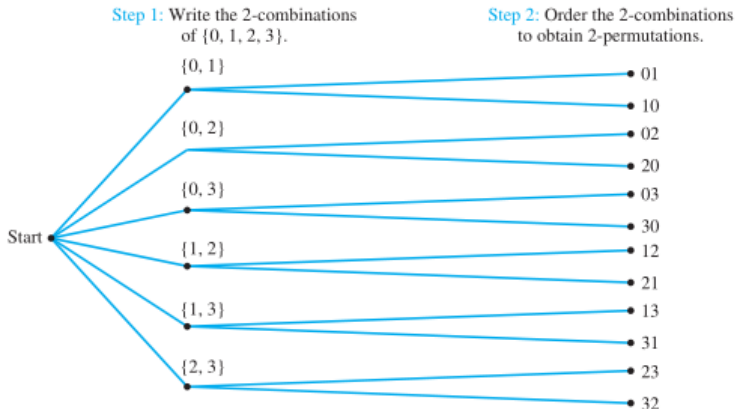
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How many unordered selections of two elements can be made from the set $\{0, 1, 2, 3\}$?

$\{0, 1\}, \{0, 2\}, \{0, 3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}$

Write all 2-permutations of the set $\{0, 1, 2, 3\}$?



Combination

- 1 Combine r objects out of n objects
- 2 Order/position does not matter

Permutation

- 1 Combine r objects out of n objects
- 2 Arrange r objects in order

“How many ways are there to arrange 5 boys and 3 girls in a row so that no two girls are adjacent?”

Many counting problems can be solved by finding the number of ways to arrange a specified number of distinct elements of a set of a particular size, where the **order of these elements matters**.

Many other counting problems can be solved by finding the number of ways to select a particular number of elements from a set of a particular size, where the **order of the elements selected does not matter**.

For example, in how many ways can we select three students from a group of five students to stand in line for a picture? How many different committees of three students can be formed from a group of four students?

Permutations	Combinations
In how many ways can we select three students from a group of five students to stand in line for a picture?	How many different committees of three students can be formed from a group of four students?
The order in which students are selected matters. There are five ways to select the first student to stand at the start of the line. Once this student has been selected, there are four ways to select the second student in the line. After the first and second students have been selected, there are three ways to select the third student in the line. By the product rule, there are $5 \cdot 4 \cdot 3 = 60$ ways to select three students from a group of five students to stand in line for a picture.	We need only find the number of subsets with three elements from the set containing the four students. This means that there are four ways to choose the three students for the committee, where the order in which these students are chosen does not matter.

Permutations	Combinations
A permutation of a set of distinct objects is an ordered arrangement of these objects. We also are interested in ordered arrangements of some of the elements of a set. An ordered arrangement of r elements of a set is called an r-permutation.	An r-combination of elements of a set is an un-ordered selection of r elements from the set. Thus, an r-combination is simply a subset of the set with r elements.
If n is a positive integer and r is an integer with $1 \leq r \leq n$, then there are $P_r^n = P(n, r) = n(n-1)(n-2) \cdots (n-r+1)$ r-permutations of a set with n distinct elements.	$P_r^n = C_r^n \times r!$

Permutations	Combinations
If n and r are integers with $0 \leq r \leq n$, then $P(n, r) = \frac{n!}{(n - r)!}$	The number of r-combinations of a set with n elements, where n is a nonnegative integer and r is an integer with $0 \leq r \leq n$, equals $C(n, r) = \frac{n!}{r!(n - r)!}.$

Example 1.3.1. Let $A = \{a, b, c, d\}$. There are altogether $P_3^4 (= 24)$ 3-permutations of A and they are shown in Example 1.2.1. These 24 3-permutations are re-grouped into 8 subsets as shown below:

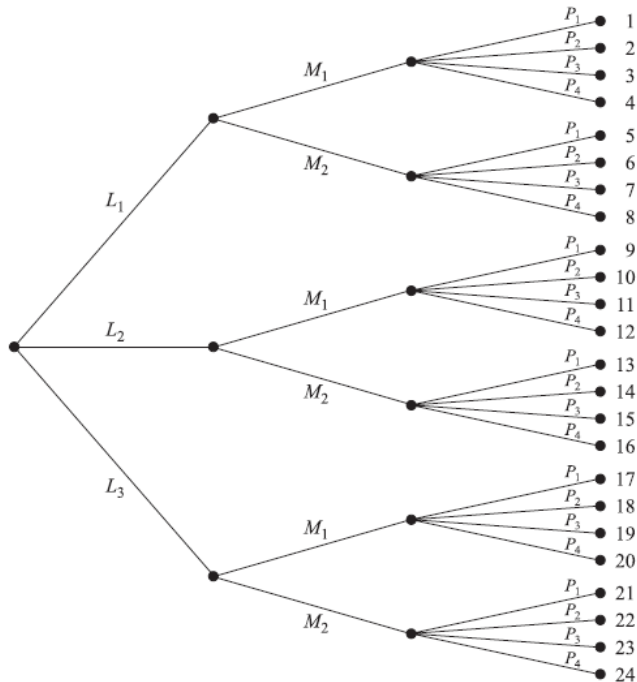
abc cab bca	acb bac cba
abd dab bda	adb bad dba
acd dac cda	adc cad dca
bcd dbc cdb	bdc cbd dcb

A typical PIN (personal identification number) is a sequence of any four symbols chosen from the 26 letters in the alphabet and the ten digits, with repetition allowed. How many different PINs are possible?

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Now suppose that repetition is not allowed.

- ① How many ways are there to select a first-prize winner, a second-prize winner, and a third-prize winner from 100 different people who have entered a contest?
- ② In how many ways can 5 differently colored marbles be arranged in a row?
- ③ Suppose that a saleswoman has to visit eight different cities. She must begin her trip in a specified city, but she can visit the other seven cities in any order she wishes. How many possible orders can the saleswoman use when visiting these cities?
- ④ How many permutations of the letters ABCDEFGH contain the string ABC ?
- ⑤ How many ways are there to select five players from a 10-member tennis team to make a trip to a match at another school?
- ⑥ How many eight-bit strings have exactly three 1's?
- ⑦ How many bit strings of length n contain exactly r 1s?

A committee of 3 members is to be formed consisting of one representative each from labor, management, and the public. If there are 3 possible representatives from labor, 2 from management, and 4 from the public, determine how many different committees can be formed using (a) the fundamental principle of counting and (b) a tree diagram.



Suppose the group of twelve consists of five men and seven women.

- ① How many five-person teams can be chosen that consist of three men and two women?
- ② How many five-person teams contain at least one man?
- ③ How many five-person teams contain at most one man?

(a) To answer this question, think of forming a team as a two-step process:

Step 1: Choose the men.

Step 2: Choose the women.

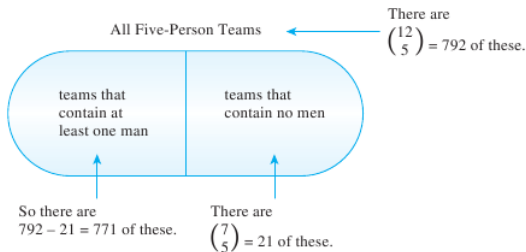
There are $\binom{5}{3}$ ways to choose the three men out of the five and $\binom{7}{2}$ ways to choose the two women out of the seven. Hence, by the product rule, $\binom{5}{3} \cdot \binom{7}{2} = 210$

How many five-person teams contain at least one man?

Observe that the set of five-person teams containing at least one man equals the set difference between the set of all five-person teams and the set of five-person teams that do not contain any men.

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Alternative way

There are

$\binom{5}{1} \binom{7}{4}$ teams with one man and four women

$\binom{5}{2} \binom{7}{3}$ teams with two men and three women

$\binom{5}{3} \binom{7}{2}$ teams with three men and two women

$\binom{5}{4} \binom{7}{1}$ teams with four men and one woman

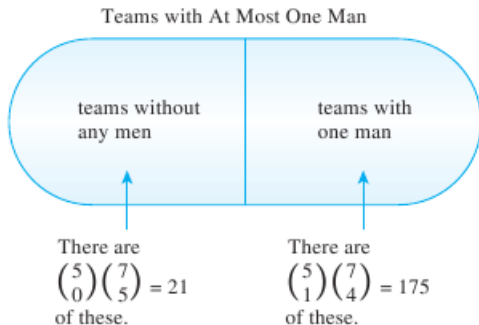
$\binom{5}{5} \binom{7}{0}$ teams with five men and no women.

Hence, by the addition rule,

$$\begin{aligned} & \left[\begin{array}{l} \text{number of teams with} \\ \text{at least one man} \end{array} \right] \\ &= \binom{5}{1} \binom{7}{4} + \binom{5}{2} \binom{7}{3} + \binom{5}{3} \binom{7}{2} + \binom{5}{4} \binom{7}{1} + \binom{5}{5} \binom{7}{0} \end{aligned}$$

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So the total number of teams with at most one man is $21 + 175 = 196$.

Permutations with Indistinguishable Objects

How many different strings can be made by reordering the letters of the word SUCCESS?

Because some of the letters of SUCCESS are the same, the answer is not given by the number of permutations of seven letters. This word contains **three Ss, two Cs, one U, and one E.**

To determine the number of different strings that can be made by reordering the letters, **first note that the three Ss can be placed among the seven positions in $C(7, 3)$ different ways, leaving four positions free.** Then the two Cs can be placed in $C(4, 2)$ ways, leaving two free positions. **The U can be placed in $C(2, 1)$ ways, leaving just one position free.** Hence E can be placed in $C(1, 1)$ way. Consequently, from the product rule, the number of different strings that can be made is

$$C(7, 3)C(4, 2)C(2, 1)C(1, 1) = \frac{7!}{3!.4!} \cdot \frac{4!}{2!.2!} \cdot \frac{2!}{1!.1!} \cdot \frac{1!}{1!0!} = \frac{7!}{3!2!1!1!} = 420$$

Consider various ways of ordering the letters in the word MISSISSIPPI: IIMSSPISSIP, ISSSPMIIPIS, PIMISSSSIIIP, and so on. How many distinguishable orderings are there?

Step 1: Choose a subset of four positions for the S 's.

Step 2: Choose a subset of four positions for the I 's.

Step 3: Choose a subset of two positions for the P 's.

Step 4: Choose a subset of one position for the M .

1 2 3 4 5 6 7 8 9 10 11

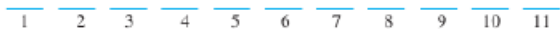
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Step 3: Choose a subset of two positions for the *P*'s.

Step 4: Choose a subset of one position for the *M*.



Since there are 11 positions in all, there are $\binom{11}{4}$ subsets of four positions for the *S*'s. Once the four *S*'s are in place, there are seven positions that remain empty, so there are $\binom{7}{4}$ subsets of four positions for the *I*'s. After the *I*'s are in place, there are three positions left empty, so there are $\binom{3}{2}$ subsets of two positions for the *P*'s. That leaves just one position for the *M*. But $1 = \binom{1}{1}$. Hence by the multiplication rule,

$$[\text{No. of ways to position all letters}] = C_4^{11} C_4^7 C_2^3 C_1^1 = \frac{11!}{4!4!2!1!} = 34650.$$

Theorem

Suppose a collection consists of n objects of which

n_1 are of type 1 and are indistinguishable from each other

n_2 are of type 2 and are indistinguishable from each other

$\dots$

$\dots$

$\dots$

n_k are of type k and are indistinguishable from each other,

and suppose that $n_1 + n_2 + \dots + n_k = n$. Then the number of distinguishable permutations of the n objects is

$$\binom{n}{n_1} \binom{n-n_1}{n_2} \binom{n-n_1-n_2}{n_3} \dots \binom{n-n_1-n_2-\dots-n_{k-1}}{n_k} = \frac{n!}{n_1! n_2! n_3! \dots n_k!}.$$

How many ways are there to distribute hands of 5 cards to each of four players from the standard deck of 52 cards?

We will use the product rule to solve this problem. To begin, note that the first player can be dealt 5 cards in $C(52, 5)$ ways. The second player can be dealt 5 cards in $C(47, 5)$ ways, because only 47 cards are left. The third player can be dealt 5 cards in $C(42, 5)$ ways. Finally, the fourth player can be dealt 5 cards in $C(37, 5)$ ways. Hence, the total number of ways to deal four players 5 cards each is

$$C(52, 5)C(47, 5)C(42, 5)C(37, 5) = \frac{52!}{5!5!5!5!32!}.$$

A box contains 24 transistors, 4 of which are defective. If 4 are sold at random, find the following probabilities.

- a. Exactly 2 are defective.
- b. None is defective.
- c. All are defective.
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(a) ${}^{24}C_2$ is wrong answer.

There are two types of transistors "Defective" and "Non-defective". We have to choose exactly 2 defective out of 4 defective transistors.

(b)

(c)

(d)

In how many ways can 10 objects be split into two groups containing 4 and 6 objects, respectively?